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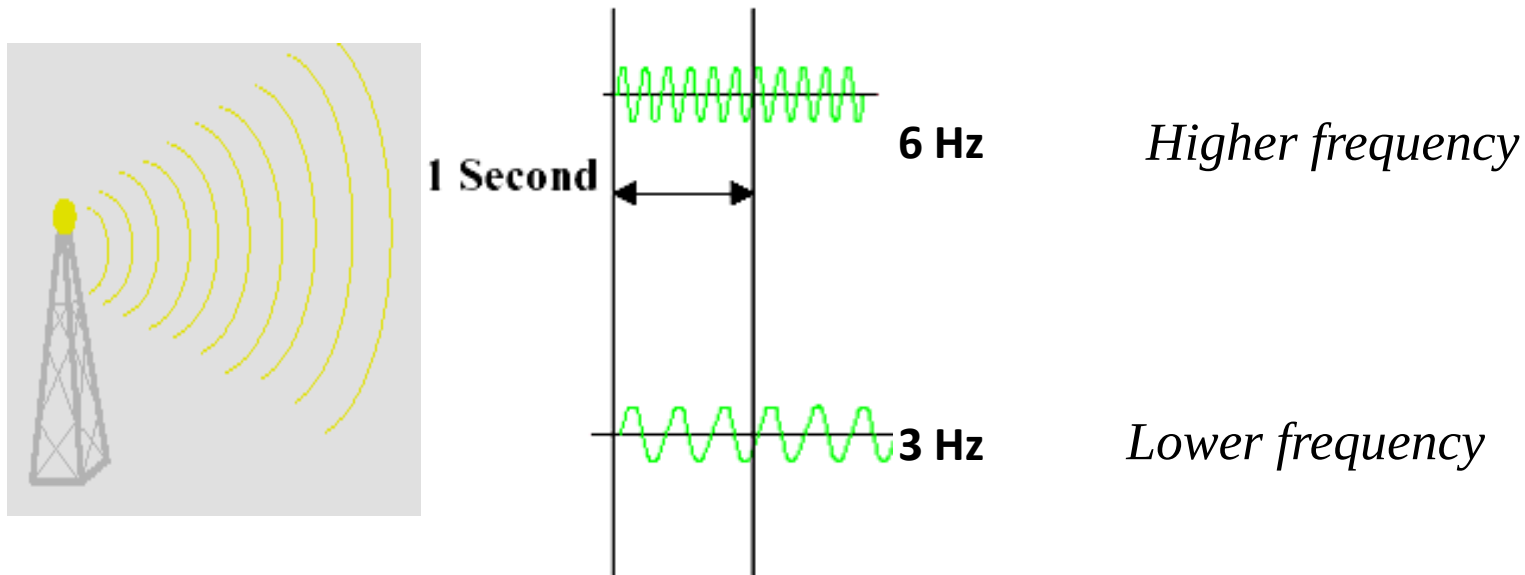
Learning Objectives of This Lecture

1. Understand the background, process, and new applications of the **frequency-domain analysis** method.
2. Proficiently master the **Fourier series analysis** methods for **periodic signals** in trigonometric and exponential forms.
3. Proficiently use the relationship between waveform symmetry and harmonic characteristics to obtain Fourier series.
4. Be familiar with the characteristics of Fourier series of typical periodic signals such as **rectangular waves**.

3.1 Introduction



- **Time-domain analysis:** The independent variable of the signal or system model is time (Chapter 2).
- **Frequency-domain analysis:** The independent variable is frequency (Chapter 3).
- **Complex frequency-domain analysis:** The independent variable is complex frequency-its real part is frequency (Chapter 4).
- **Frequency** is the number of periodic changes completed per unit time (unit: cycles per second or **Hertz Hz**), which describes the speed of signal change.
- The frequency of a sinusoidal signal can be directly observed from the time domain.



3.1 Introduction



- In 1807, *Joseph Fourier*, the French mathematician and physicist, proposed the principle of **expanding periodic functions into a series of sine and cosine waves**, while studying heat conduction theory, laying the foundation for frequency-domain analysis.
- *Siméon Poisson*, *Carl Gauss* and other scientists applied this achievement to electrical engineering.
- In the late 19th century, capacitors for practical engineering applications were invented.
- Since the 20th century, the solution to a series of specific problems such as resonant circuits, filters, and sinusoidal oscillators opened up a broad prospect for the further application of sinusoidal functions and Fourier analysis.
- The **Fourier Transform** has many advantages in the theoretical research and engineering application of electrical, communication and control systems.

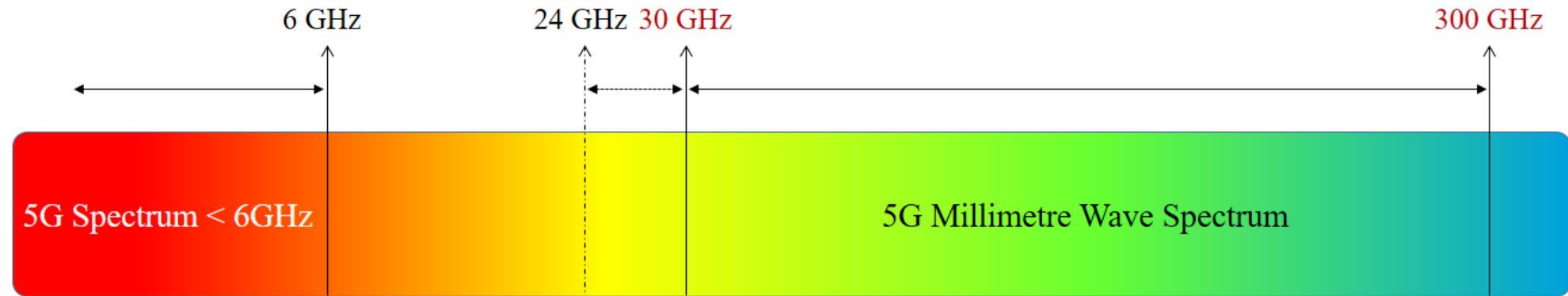


Joseph Fourier
(1768-1830)

3.1 Introduction



Fourier analysis serves as the basis of spectrum allocation.



Frequencies used in radio communication											
3–30 Hz	30–300 Hz	300–3000 Hz	3–30 kHz	30–300 kHz	300 kHz – 3 MHz	3–30 MHz	30–300 MHz	300 MHz – 3 GHz	3–30 GHz	30–300 GHz	300 GHz – 3 THz
10^5 – 10^4 km	10^4 – 10^3 km	10^3 –100 km	100–10 km	10–1 km	1 km – 100 m	100–10 m	10–1 m	1 m – 10 cm	10–1 cm	1 cm – 1 mm	1 mm – 0.1 mm
ELF	SLF	ULF	VLF	LF	MF	HF	VHF	UHF	SHF	EHF	THF

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- Frequency-domain analysis reveals the inherent frequency characteristics of signals and the close relationship between the time characteristics and frequency characteristics of signals, thus deriving important concepts such as signal spectrum, bandwidth, filtering, modulation, and frequency division multiplexing.
- Frequency-domain analysis is carried out through Fourier transform (Fourier analysis). The signal is decomposed orthogonally, that is, decomposed into a combination of trigonometric functions or complex exponential functions.

Fourier Analysis

Fourier series expansion: **Periodic signals** can be expressed as a **weighted sum** of **sinusoidal signals with harmonic relationships**.

Fourier transform: **Aperiodic signals** can be expressed as a **weighted integral** of **sinusoidal signals over all frequency components** from 0 to infinitely high frequencies.

Any periodic signal can be decomposed into a linear superposition of sinusoidal functions. This statement is ().

- ☐ A True
- ☒ B False

提交

3.2 Fourier Series Analysis of Periodic Signals

- In 1807, *Fourier* proposed that any periodic signal could be expanded into a superposition of sinusoidal signals with harmonic relationships.
- *Lagrange* disapproved the claim because there were breakpoints in actual signals.
- In 1822, *Fourier* published *Théorie analytique de la chaleur* (*The Analytical Theory of Heat*).
- In 1829, *Dirichlet* proved that periodic signals could be expanded into Fourier series only when certain conditions are met.
- **Dirichlet conditions (sufficient but not necessary conditions for the existence of Fourier series):**
 - (1) Within one period, the signal is **continuous** or has only **a finite number of first-kind discontinuities** (the function has finite left and right limits at the discontinuity, e.g., $x(t)=|\sin(t)|/\sin(t)$ at $t=0$).
 - (2) Within one period, the signal has **a finite number of maximum and minimum values**.
 - (3) Within one period, the signal is **absolutely integrable**, i.e., $\int_0^T |f(t)| dt < \infty$.
- **Signals in engineering all satisfy Dirichlet conditions.**

3.2 Fourier Series Analysis of Periodic Signals

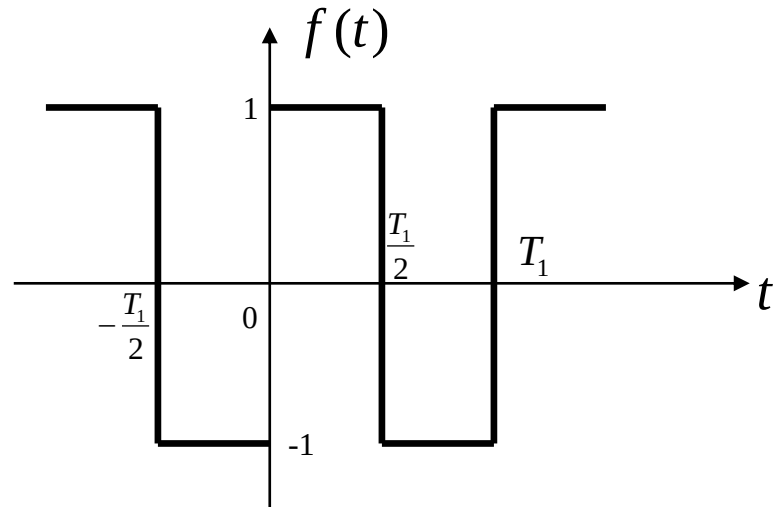


- Any periodic function that satisfies the Dirichlet conditions can be expanded into an infinite series of orthogonal functions.
- If the orthogonal function set is a **trigonometric** function set or a **complex exponential** function set, the series expanded by the periodic function is called "Fourier series".
- Why choose trigonometric functions (**sinusoidal waves**) as the basic signals of Fourier series?
 - **Sinusoidal waves have fidelity**: After a sinusoidal signal passes through an LTI system, the output is still a sinusoidal signal—only the amplitude and phase may change, while the frequency and waveform remain unchanged.
 - **The integral and derivative of a trigonometric (or exponential) function are still trigonometric (or exponential) functions.**

3.2 Fourier Series Analysis of Periodic Signals

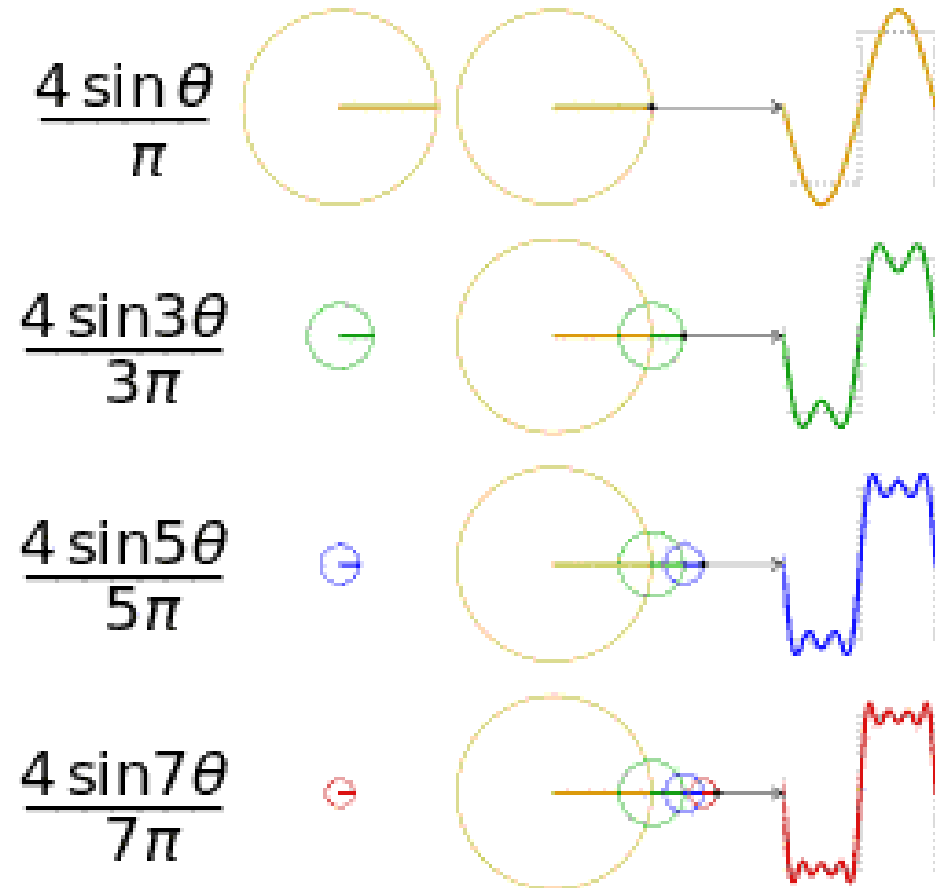


Generation of Periodic Rectangular Signals



How many sinusoidal waves need to be superimposed to form a rectangular signal?

Answer: Infinite.



3.2 Fourier Series Analysis of Periodic Signals



3.2.1 Fourier Series in Trigonometric Form

Let $f(t)$ be a periodic signal with period T_1 .

Its **fundamental angular frequency** is $\omega_1 = 2\pi f_1 = \frac{2\pi}{T_1}$.

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

DC component: $a_0 = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) dt$

Amplitude of cosine component : $a_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \cos n\omega_1 t dt$

Amplitude of sine component: $b_n = \frac{2}{T_1} \int_{t_0}^{t_0+T_1} f(t) \sin n\omega_1 t dt$

The integral limits in the above formulas are generally set as: $0 \sim T_1$ or $-\frac{T_1}{2} \sim \frac{T_1}{2}$

3.2 Fourier Series Analysis of Periodic Signals



The Fourier series in trigonometric form can also be expressed as:

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \quad (2)$$

where $c_0 = a_0$ $c_n^2 = a_n^2 + b_n^2$ $\varphi_n = \arctan(-\frac{b_n}{a_n})$

Proof:

$$\begin{aligned} f(t) &= c_0 + \sum_{n=1}^{\infty} c_n \left[\frac{a_n}{c_n} \cos(n\omega_1 t) + \frac{b_n}{c_n} \sin(n\omega_1 t) \right] \\ &= c_0 + \sum_{n=1}^{\infty} c_n [\cos(n\omega_1 t) \cos(-\varphi_n) + \sin(n\omega_1 t) \sin(-\varphi_n)] \\ &= c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \end{aligned}$$

3.2 Fourier Series Analysis of Periodic Signals



3.2.2 Fourier Series in Exponential Form

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

Based on Euler's formula,

$$\cos(n\omega_1 t) = \frac{1}{2}(e^{jn\omega_1 t} + e^{-jn\omega_1 t}), \quad \sin(n\omega_1 t) = \frac{1}{2j}(e^{jn\omega_1 t} - e^{-jn\omega_1 t})$$

Substitute into (1),

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n - jb_n}{2} e^{jn\omega_1 t} + \frac{a_n + jb_n}{2} e^{-jn\omega_1 t} \right)$$

$$\text{Let } F(n\omega_1) = \frac{1}{2}(a_n - jb_n)$$

$$F(-n\omega_1) = \frac{1}{2}(a_n + jb_n)$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} F(n\omega_1) e^{jn\omega_1 t} + \sum_{n=1}^{\infty} F(-n\omega_1) e^{-jn\omega_1 t} = \sum_{n=-\infty}^{\infty} F(n\omega_1) e^{jn\omega_1 t}$$

3.2 Fourier Series Analysis of Periodic Signals



$$f(t) = \sum_{n=-\infty}^{\infty} F(n\omega_1) e^{jn\omega_1 t}$$

Exponential Form:

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (3)$$

where $F_n = \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jn\omega_1 t} dt$ -----Complex amplitude

$$F_0 = a_0 = c_0$$

$$F_n = |F_n| e^{j\varphi_n} = \frac{1}{2} (a_n - jb_n)$$

$$|F_n| = \frac{1}{2} \sqrt{a_n^2 + b_n^2} = \frac{1}{2} c_n \quad \varphi_n = \arctan \left(-\frac{b_n}{a_n} \right)$$

3.2 Fourier Series Analysis of Periodic Signals



3.2.3 Spectrum of Periodic Signals and Its Characteristics

1. Spectrum of Periodic Signals

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_1 t + b_n \sin n\omega_1 t) \quad (1)$$

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_1 t + \varphi_n) \quad (2)$$

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_1 t} \quad (3)$$

To clearly indicate which frequency components a signal contains and the proportion of each component, a **spectrum diagram** can be drawn for intuitive representation.

If frequency is taken as the horizontal axis and amplitude/phase as the vertical axis, the relative magnitude and phase of each frequency component can be observed directly. Such diagrams are called the **amplitude spectrum** and **phase spectrum** of the signal in trigonometric form.

3.2 Fourier Series Analysis of Periodic Signals



Spectrum Diagram of Periodic Rectangular Signals



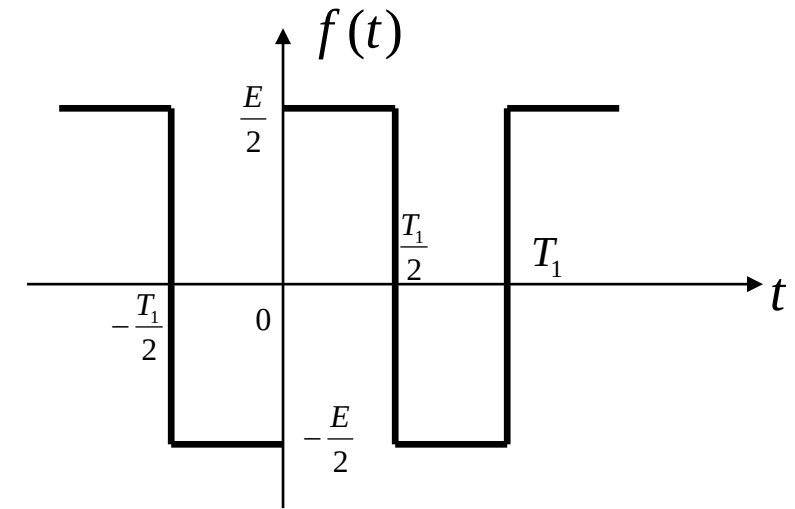
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3.2 Fourier Series Analysis of Periodic Signals

Example 3-1: Find the Fourier series in trigonometric and exponential forms of the periodic rectangular signal shown in the figure, and draw their respective spectrum diagrams.

Solution: The expression of $f(t)$ within one period is:

$$f(t) = \begin{cases} \frac{E}{2} & 0 \leq t < \frac{T_1}{2} \\ -\frac{E}{2} & -\frac{T_1}{2} \leq t < 0 \end{cases} \quad (\text{Odd function})$$



$$a_0 = \frac{1}{T_1} \int_{-T_1/2}^{T_1/2} f(t) dt = 0 \quad a_n = \frac{2}{T_1} \int_{-T_1/2}^{T_1/2} f(t) \cos n\omega_1 t dt = 0$$

$$b_n = \frac{2}{T_1} \int_{-T_1/2}^{T_1/2} f(t) \sin n\omega_1 t dt = \frac{E}{n\pi} [1 - \cos(n\pi)] = \begin{cases} \frac{2E}{n\pi} & n = 1, 3, 5 \dots \\ 0 & n = 2, 4, 6 \dots \end{cases}$$

3.2 Fourier Series Analysis of Periodic Signals



$$c_0 = 0$$

$$c_n = b_n = \begin{cases} \frac{2E}{n\pi} & n = 1, 3, 5 \dots \\ 0 & n = 2, 4, 6 \dots \end{cases}$$

$$\varphi_n = \arctan\left(-\frac{b_n}{a_n}\right) = -\frac{\pi}{2} \quad (n = 1, 3, 5 \dots)$$

So,
$$f(t) = \frac{2E}{\pi} \sum_{n=1,3,5\dots}^{\infty} \frac{1}{n} \sin n\omega_1 t$$
$$= \frac{2E}{\pi} \left(\sin \omega_1 t + \frac{1}{3} \sin 3\omega_1 t + \frac{1}{5} \sin 5\omega_1 t + \dots \right)$$

or
$$f(t) = \frac{2E}{\pi} \sum_{n=1,3,5\dots}^{\infty} \frac{1}{n} \cos\left(n\omega_1 t - \frac{\pi}{2}\right)$$

3.2 Fourier Series Analysis of Periodic Signals



$$F_n = \frac{1}{2}(a_n - jb_n) = -j\frac{b_n}{2} = \begin{cases} -\frac{jE}{n\pi} & n = \pm 1, \pm 3, \pm 5 \dots \\ 0 & n = \pm 2, \pm 4, \pm 6 \dots \end{cases}$$

$$f(t) = \left(\frac{jE}{\pi} e^{-j\omega_1 t} - \frac{jE}{\pi} e^{j\omega_1 t} \right) + \left(\frac{jE}{3\pi} e^{-j3\omega_1 t} - \frac{jE}{3\pi} e^{j3\omega_1 t} \right) + \dots$$

$$|F_n| = \frac{E}{|n|\pi} \quad (n = \pm 1, \pm 3, \pm 5 \dots)$$

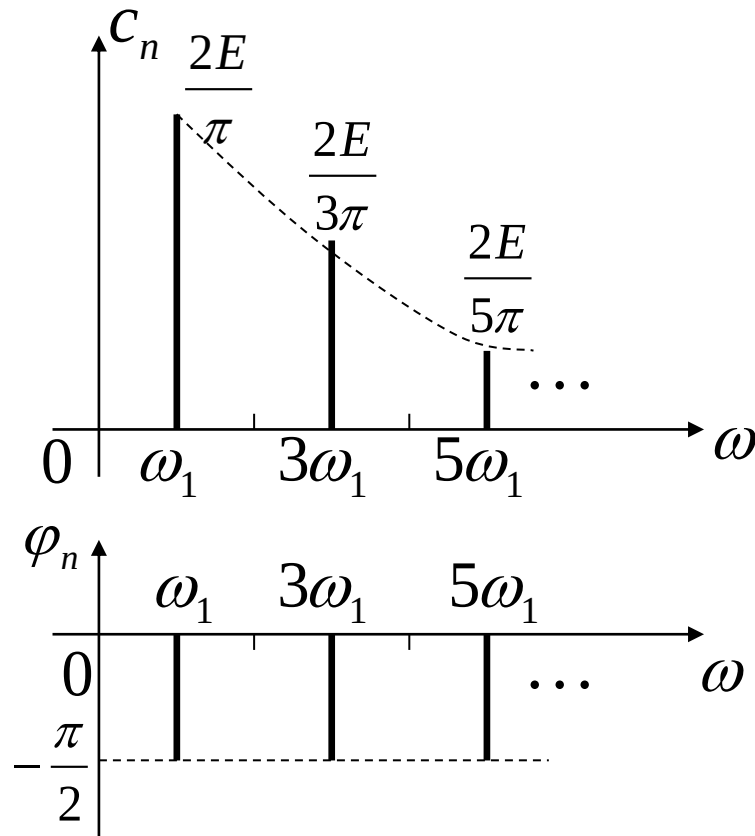
$$\varphi_n = \begin{cases} -\frac{\pi}{2} & (n = 1, 3, 5 \dots) \\ \frac{\pi}{2} & (n = -1, -3, -5 \dots) \end{cases}$$

3.2 Fourier Series Analysis of Periodic Signals



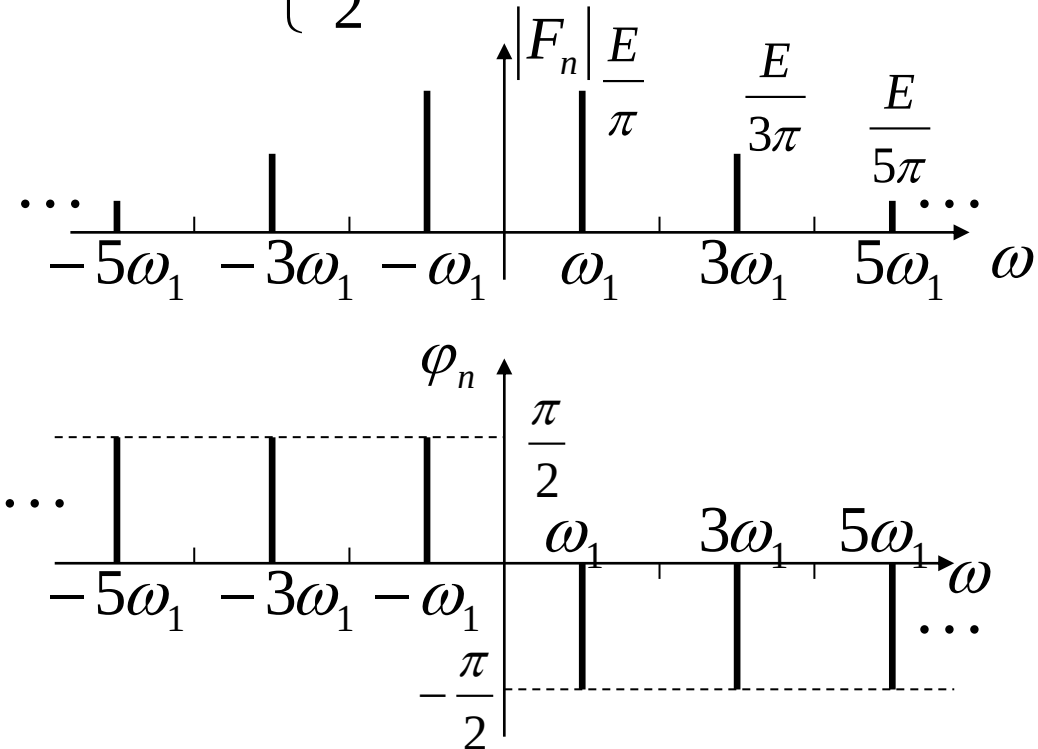
$$c_n = \begin{cases} \frac{2E}{n\pi} & n = 1, 3, 5 \dots \\ 0 & n = 2, 4, 6 \dots \end{cases}$$

$$\varphi_n = -\frac{\pi}{2} \quad (n = 1, 3, 5 \dots)$$



$$|F_n| = \frac{E}{|n|\pi} \quad (n = \pm 1, \pm 3, \pm 5 \dots)$$

$$\varphi_n = \begin{cases} -\frac{\pi}{2} & (n = 1, 3, 5 \dots) \\ \frac{\pi}{2} & (n = -1, -3, -5 \dots) \end{cases}$$



2. Characteristics of Periodic Signal Spectrum

- (1) **Discreteness:** The spectrum is discrete (not continuous), which is called a discrete spectrum.
- (2) **Harmonicity:** Spectral lines appear at integer multiples of the fundamental frequency ω_1 .
- (3) **Convergence:** The amplitude of spectral lines in the amplitude spectrum gradually decays to zero as n increases ($n \rightarrow \infty$).

3.2 Fourier Series Analysis of Periodic Signals

3.2.4 Relationship Between Waveform Symmetry and Harmonic Characteristics

When a signal $f(t)$ is expanded into a Fourier series, if $f(t)$ is a real function and its waveform satisfies a certain **symmetry**, some terms in the Fourier series disappear, and the expressions of the remaining coefficients become simpler.

There are two types of waveform symmetry:

- (1) **symmetry about the entire period;**
- (2) **symmetry about half the period.**

3.2 Fourier Series Analysis of Periodic Signals

1. Even function

$$f(t) = f(-t)$$

$$b_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \sin n\omega_1 t dt = 0$$

$$a_0 = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) dt = \frac{2}{T_1} \int_0^{\frac{T_1}{2}} f(t) dt$$

$$a_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt$$

The Fourier series of an even function **does not contain sine components**, but may contain DC and cosine components.

2. Odd Function

$$f(t) = -f(-t)$$

$$a_0 = \frac{1}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) dt = 0$$

$$a_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = 0$$

$$b_n = \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \sin n\omega_1 t dt = \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \sin n\omega_1 t dt$$

The Fourier series of an odd function **does not contain DC and cosine components**, but only sine components.

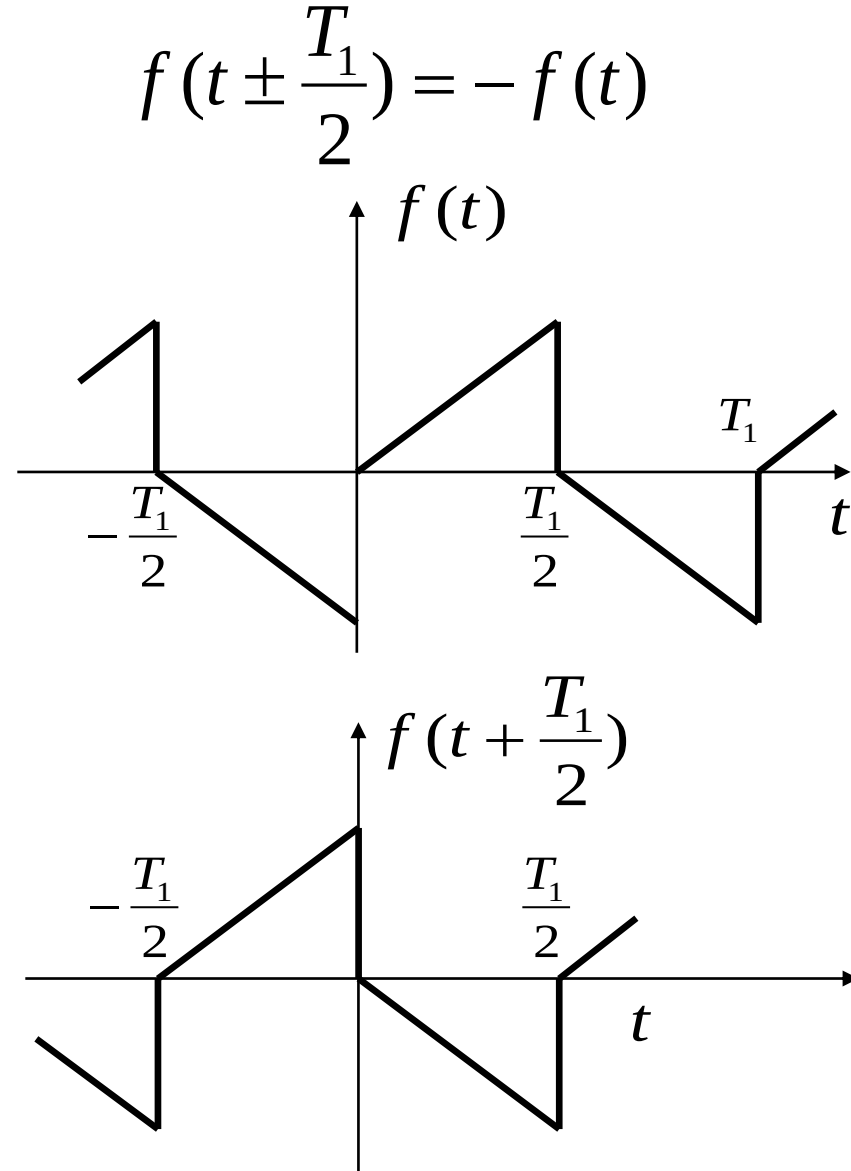
After adding a DC component to an odd function, the Fourier series still contains sine components. This statement is ().

☒ A True

☐ B False

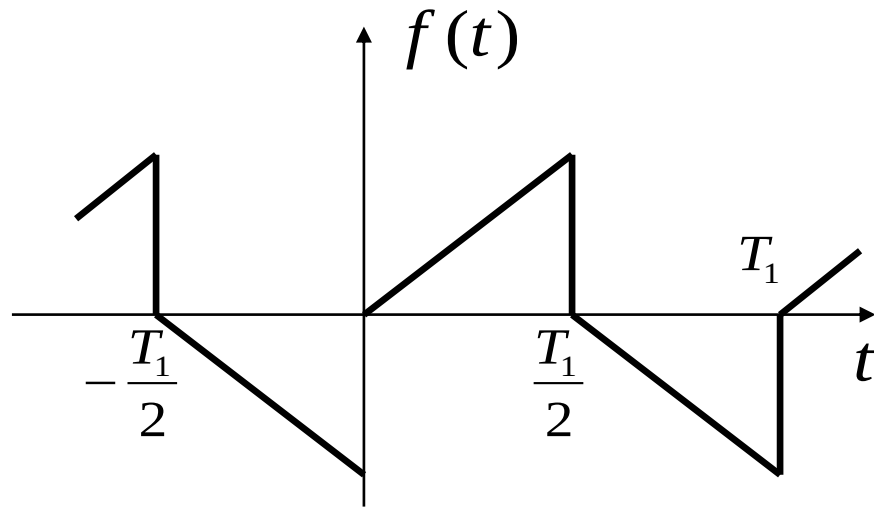
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3. Odd Harmonic Function (Half-Wave Odd Symmetry)



$$\begin{aligned}
 a_n &= \frac{2}{T_1} \int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt \\
 &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt + \int_{-\frac{T_1}{2}}^0 f(t) \cos(n\omega_1 t) dt \right] \\
 &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt - \int_{-\frac{T_1}{2}}^0 f(t + \frac{T_1}{2}) \cos(n\omega_1 t) dt \right] \\
 &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt - \int_0^{\frac{T_1}{2}} f(\tau) \cos[n\omega_1(\tau - T_1/2)] d\tau \right] \\
 &= \frac{2}{T_1} \left[\int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t) dt - \int_0^{\frac{T_1}{2}} f(t) \cos(n\omega_1 t - n\pi) dt \right] \\
 &= \begin{cases} 0 & (n = 0, 2, 4, 6 \dots) \\ \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt & (n = 1, 3, 5 \dots) \end{cases}
 \end{aligned}$$

3.2 Fourier Series Analysis of Periodic Signals



$$a_0 = 0$$

$$a_n = \begin{cases} 0 & (n = 2, 4, 6 \dots) \\ \frac{4}{T_1} \int_0^{T_1/2} f(t) \cos n\omega_1 t dt & (n = 1, 3, 5 \dots) \end{cases}$$

$$b_n = \begin{cases} 0 & (n = 2, 4, 6 \dots) \\ \frac{4}{T_1} \int_0^{T_1/2} f(t) \sin n\omega_1 t dt & (n = 1, 3, 5 \dots) \end{cases}$$

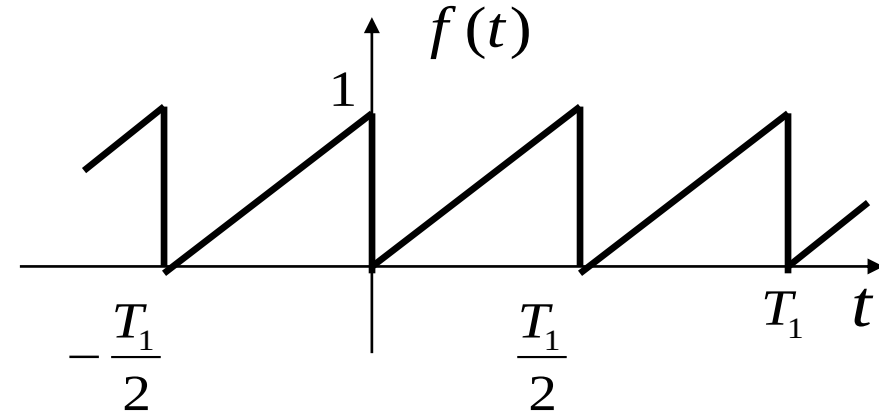
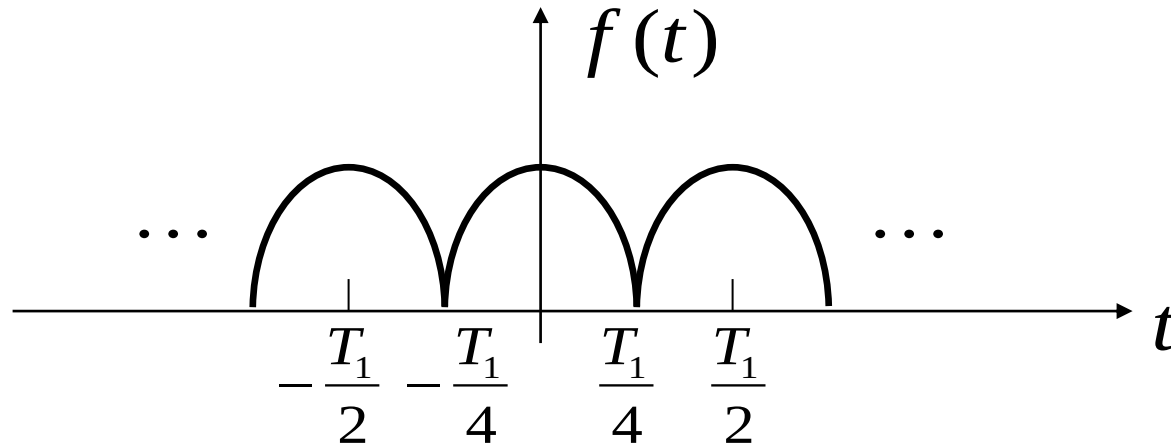
The Fourier series of a half-wave odd symmetric function contains only sine and cosine components of the fundamental wave and odd harmonics, and **does not contain DC components and even harmonic components**—thus it is called an **odd harmonic function**.

3.2 Fourier Series Analysis of Periodic Signals



4. Even Harmonic Function (Half-Wave Even Symmetry)

$$f(t \pm \frac{T_1}{2}) = f(t)$$



The Fourier series of a half-wave even symmetric function contains only sine and cosine components of even harmonics (as well as DC components), and **does not contain odd harmonic components**—thus it is called an **even harmonic function**.

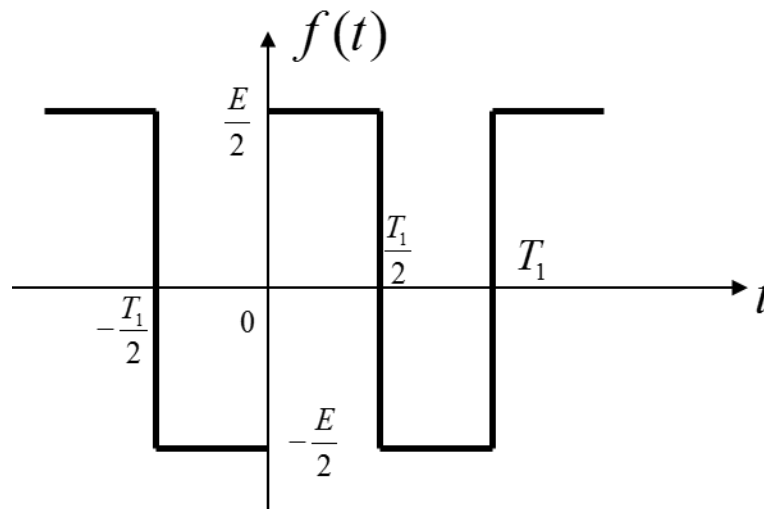
The zero components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are ().

☒ A $a_n = 0 \quad (n = 0, 2, 4 \cdots)$

☒ B $a_n = 0 \quad (n = 1, 3, 5 \cdots)$

☐ C $b_n = 0 \quad (n = 1, 3, 5 \cdots)$

☒ D $b_n = 0 \quad (n = 2, 4, 6 \cdots)$



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3.2 Fourier Series Analysis of Periodic Signals



Which components of the Fourier series in trigonometric form for the periodic rectangular signal shown in the figure are zero?

This is an **odd function**, so the DC component and cosine components are zero.

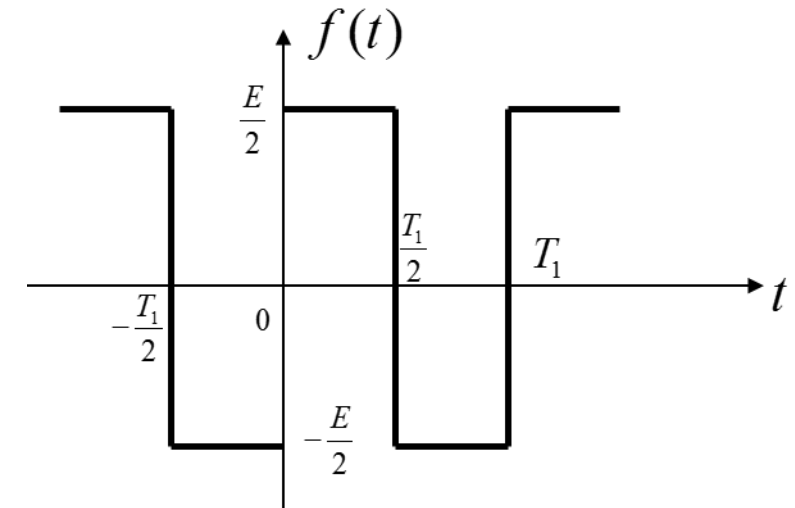
$$a_n = 0 \quad (n = 0, 2, 4 \dots)$$

$$a_n = 0 \quad (n = 1, 3, 5 \dots)$$

It is also an **odd harmonic function** (half-wave odd symmetry), so the DC and even harmonic components are zero.

$$a_n = 0 \quad (n = 0, 2, 4 \dots)$$

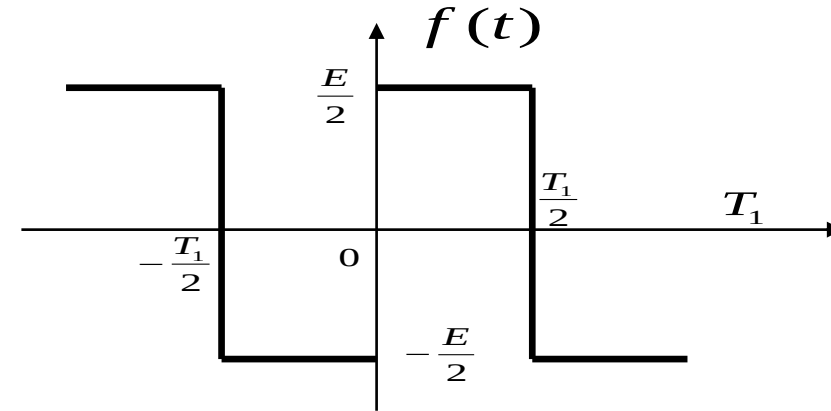
$$b_n = 0 \quad (n = 2, 4, 6 \dots)$$



3.2 Fourier Series Analysis of Periodic Signals



3.2.5 Gibbs Phenomenon



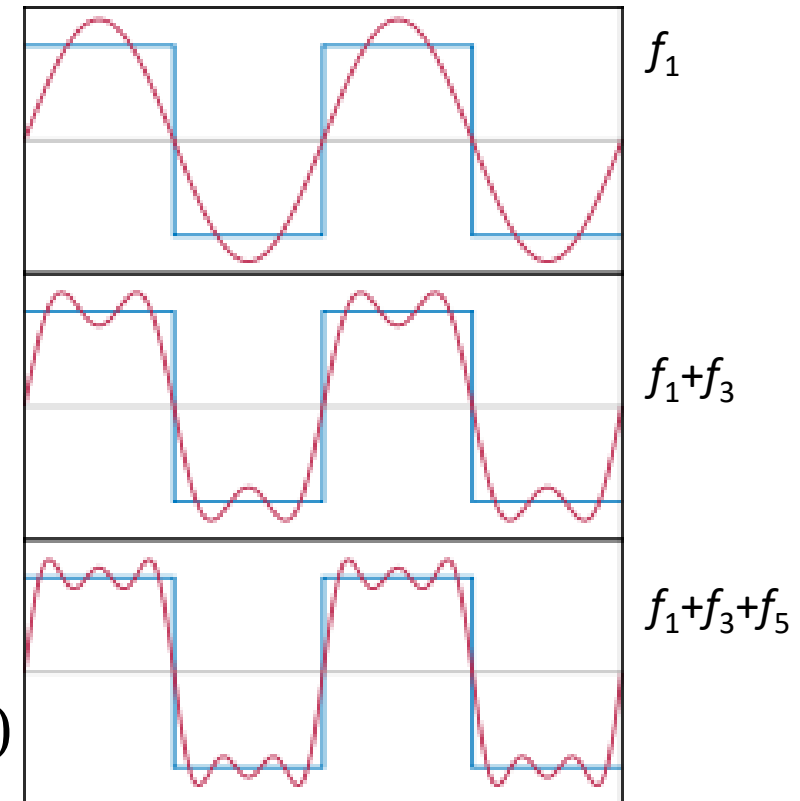
$n=1:$ $f(t) \approx \frac{2E}{\pi} \sin \omega_1 t$

$n=3:$

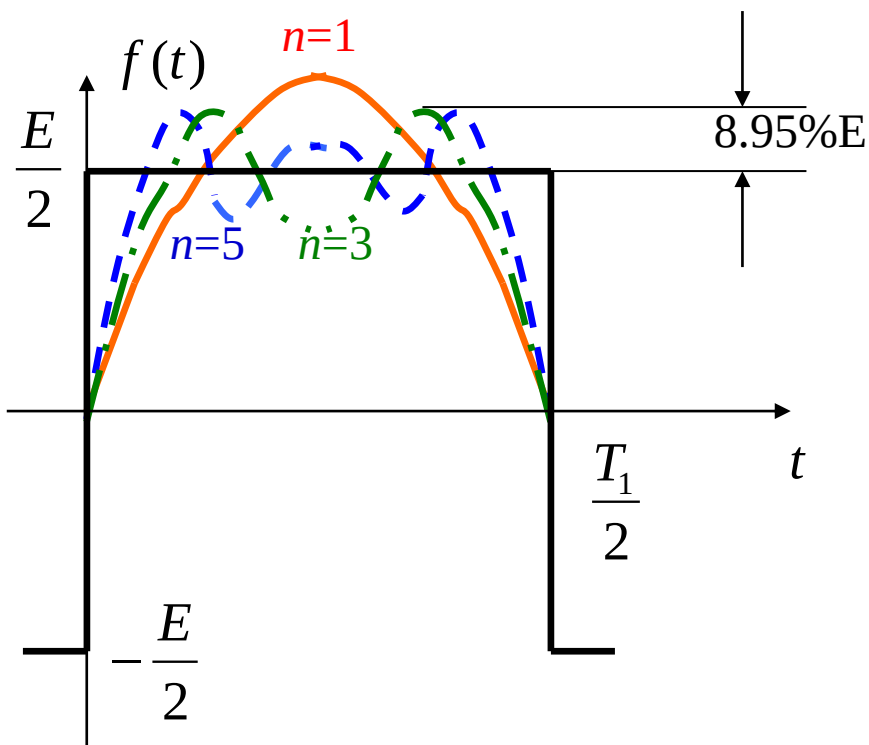
$$f(t) \approx \frac{2E}{\pi} \left(\sin \omega_1 t + \frac{1}{3} \sin 3\omega_1 t \right)$$

$n=5:$

$$f(t) \approx \frac{2E}{\pi} \left(\sin \omega_1 t + \frac{1}{3} \sin 3\omega_1 t + \frac{1}{5} \sin 5\omega_1 t \right)$$



3.2 Fourier Series Analysis of Periodic Signals



When the number of terms N is large, the peak value tends to a constant (**approximately 9% of the total jump value**) and gradually decays in the form of undulating oscillations starting from the discontinuous point.

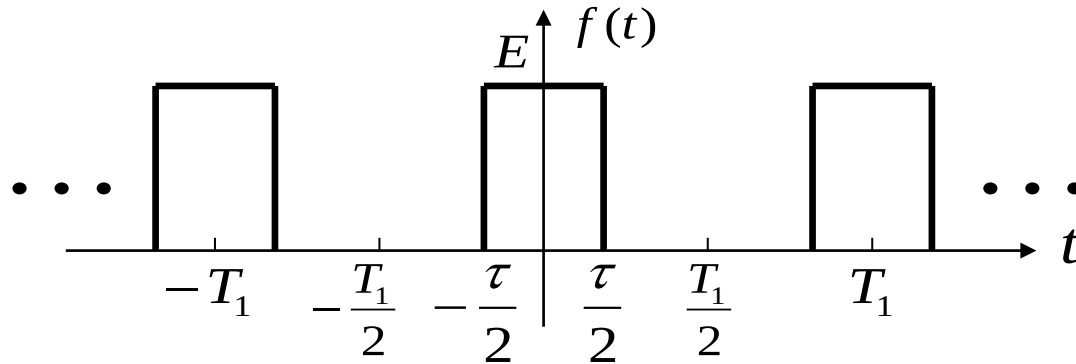
High-frequency components mainly affect the **transition edge** of the pulse.
Low-frequency components mainly affect the **top** of the pulse.

The more drastic the waveform change of $f(t)$, the richer the high-frequency components it contains; the slower the change, the richer the low-frequency components it contains.

3.3 Fourier Series of Typical Periodic Signals



Fourier Series of Periodic Rectangular Pulse Signals



$b_n = 0$ (The Fourier series of an even function does not contain sine terms)

$$a_0 = \frac{2}{T_1} \int_0^{\frac{T_1}{2}} f(t) dt = \frac{2}{T_1} \int_0^{\frac{\tau}{2}} E dt = \frac{E\tau}{T_1}$$

$$\begin{aligned} a_n &= \frac{4}{T_1} \int_0^{\frac{T_1}{2}} f(t) \cos n\omega_1 t dt = \frac{4}{T_1} \int_0^{\frac{\tau}{2}} E \cos n\omega_1 t dt = \frac{4E}{n\omega_1 T_1} \sin\left(\frac{n\omega_1 \tau}{2}\right) = \frac{2E\tau}{T_1} \cdot \frac{\sin\left(\frac{n\omega_1 \tau}{2}\right)}{\frac{n\omega_1 \tau}{2}} \\ &= \frac{2E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1 \tau}{2}\right) = c_n \end{aligned}$$

3.3 Fourier Series of Typical Periodic Signals



Fourier series of a periodic rectangular pulse signals in trigonometric form:

$$f(t) = \frac{E\tau}{T_1} + \frac{2E\tau}{T_1} \sum_{n=1}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) \cos n\omega_1 t$$

Since $F_n = \frac{1}{2}(a_n - jb_n) = \frac{1}{2}a_n = \frac{E\tau}{T_1} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right)$

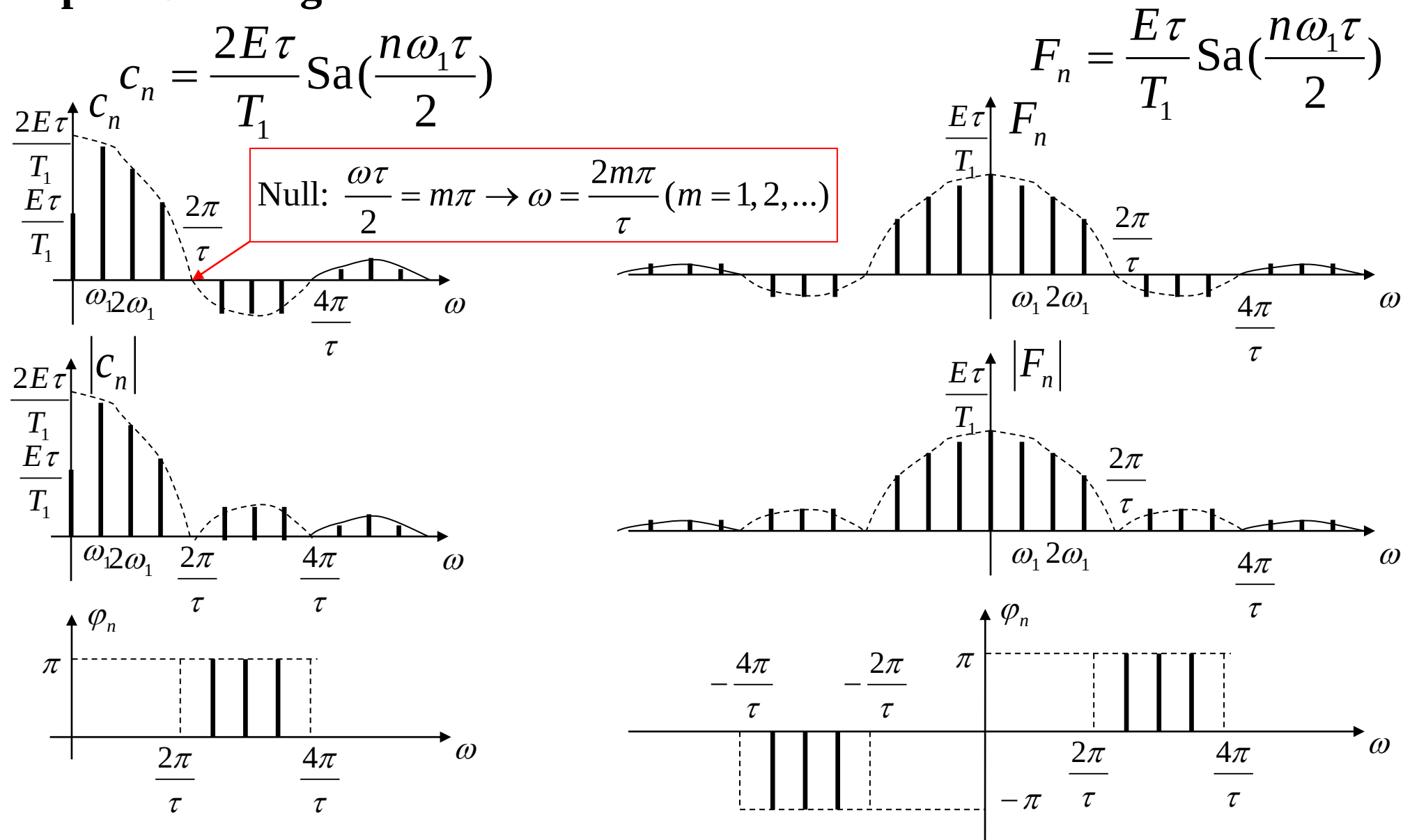
Fourier series in exponential form:

$$f(t) = \frac{E\tau}{T_1} \sum_{n=-\infty}^{\infty} \text{Sa}\left(\frac{n\omega_1\tau}{2}\right) e^{jn\omega_1 t}$$

3.3 Fourier Series of Typical Periodic Signals



Spectrum Diagram



- Repeat the examples in the lecture notes.
- 3.50(a)(d) in the textbook.